ID #1:_____
ID #2:_____

We now construct the definite integral $\int_a^b f(x)dx$. Let f be a bounded function defined on the closed interval $[a,b]$. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a *partition* of $[a,b]$, ordered so that

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

Let

$$m_i := \inf\{f(x) : x_{i-1} \leq x \leq x_i\},$$
$$M_i := \sup\{f(x) : x_{i-1} \leq x \leq x_i\},$$

for each $i = 1, \ldots, n$. Define the *upper* and *lower sums* of f for this partition as

$$U(f,P) := \sum_{i=1}^{n} M_i(x_i - x_{i-1}),$$
$$L(f,P) := \sum_{i=1}^{n} m_i(x_i - x_{i-1}),$$

respectively. We set out to define $\int_a^b f(x)dx$ by the requirement that

$$L(f,P) \leq \int_a^b f(x)dx \leq U(f,P)$$

for every partition P of $[a,b]$. We will see that this requirement uniquely characterizes $\int_a^b f(x)dx$, when it exists.

1. Let P be any partition of $[a,b]$.
   (a) Show that $m_i \leq M_i$ for each $i = 0, \ldots, n$.
   (b) Show that part (a) implies $L(f,P) \leq U(f,P)$.

We have just proved that $L(f,P) \leq U(f,P)$ for every partition P. However, something less obvious *ought* to be true: if P_1 and P_2 are *any* two partitions of $[a,b]$, then we should have

$$L(f,P_1) \leq U(f,P_2)$$

since we should have $L(f,P_1) \leq$ "area of $R(f,a,b)$" $\leq U(f,P_2)$. To prove that this is indeed true, we need to first consider how the upper and lower sums change when we increase the number of points in the partition. If P_1 and P_2 are two partitions of $[a,b]$, we say that P_2 is a *refinement* of P_1 (or that P_2 *refines* P_1) if $P_1 \subseteq P_2$. That is, P_2 is a refinement of P_1 if P_2 consists of all the points in P_1 and possibly more.

As we have seen in examples, refining a partition should give a larger underestimate and a smaller overestimate:

Lemma. (Refinement Lemma) Let $P_1 \subseteq P_2$ be partitions of $[a,b]$, such that P_2 is a refinement of P_1. Then for any bounded function $f : [a,b] \to \mathbb{R}$ we have

$$L(f,P_1) \leq L(f,P_2) \leq U(f,P_2) \leq U(f,P_1).$$

2. To prove the refinement lemma, it is sufficient to prove the special case of adding a single point u to $[a,b]$. Assume $a \leq u \leq b$ and let

$$\begin{aligned} m &= \inf\{f(x) : x \in [a,b]\},\ M = \sup\{f(x) : x \in [a,b]\} \\ m_1 &= \inf\{f(x) : x \in [u,b]\},\ M_1 = \sup\{f(x) : x \in [u,b]\} \\ m_2 &= \inf\{f(x) : x \in [a,u]\}, M_2 = \sup\{f(x) : x \in [a,u]\}. \end{aligned}$$

   (a) Show that $m \leq m_i$ and $M_i \leq M$, for $i = 1,2$.

   (b) Use your answers to 1(b) and 2(a) to show that

$$L(f,P_1) \leq L(f,P_2) \leq U(f,P_2) \leq U(f,P_1).$$

   Hint: Use the fact that $b-a = (b-u)+(u-a)$ to help you rewrite $L(f,P_1), U(f,P_1)$.

Theorem. Let P_1, P_2 be any partitions of $[a,b]$ (that is, neither is necessarily a refinement of the other), and let $f : [a,b] \to \mathbb{R}$ be a bounded function. Then

$$L(f,P_1) \leq U(f,P_2).$$

3. Use the Refinement Lemma to prove the theorem above. *Hint:* Consider the partition $P = P_1 \cup P_2$.

The theorem above shows that every upper sum is an upper bound for the set of all lower sums, and that every lower sum is a lower bound for the set of all upper sums. Since these are nonempty subsets of $\mathbb{R}$, by completeness the set of all lower sums has a supremum and the set of all upper sums has an infimum. Let

$$\underline{\int_a^b} f := \sup\{L(f,P) : P \text{ is a partition of } [a,b]\}$$
$$\overline{\int_a^b} f := \inf\{U(f,P) : P \text{ is a partition of } [a,b]\}$$

These are called the *lower* and *upper integrals* of f on $[a,b]$, respectively.

4. Show that

$$L(f,P) \leq \underline{\int_a^b} f \leq U(f,P)$$

and

$$L(f,P) \leq \overline{\int_a^b} f \leq U(f,P)$$

for every partition P of $[a,b]$.

5. Show that $\underline{\int}_a^b f \leq \overline{\int}_a^b f$. (*Hint*: Assume that $\underline{\int}_a^b f > \overline{\int}_a^b f$ and derive a contradiction by considering the midpoint of these two numbers.)

Combining the results of questions 8 and 9, we have

$$L(f,P) \leq \underline{\int}_a^b f \leq \overline{\int}_a^b f \leq U(f,P)$$

for every partition P of $[a,b]$.

6. Show that if $\underline{\int}_a^b f = \overline{\int}_a^b f$, then this is the unique number between the lower sum and upper sum for all partitions.

We set out to define $\int_a^b f(x)dx$ by the requirement that $L(f,P) \leq \int_a^b f(x)dx \leq U(f,P)$ for every partition P. We see that this is indeed the case if and only if $\underline{\int}_a^b f(x)dx = \overline{\int}_a^b f(x)dx$. If $\underline{\int}_a^b f(x)dx < \overline{\int}_a^b f(x)dx$, then every number x such that $\underline{\int}_a^b f < x < \overline{\int}_a^b f$ satisfies $L(f,P) < x < U(f,P)$ for every partition P.

Definition. Let $f : [a,b] \to \mathbb{R}$ be a bounded function. If $\underline{\int}_a^b f(x)dx = \overline{\int}_a^b f(x)dx$, then we say f is *integrable* and define the *definite integral of f on $[a,b]$* by

$$\int_a^b f(x)dx = \underline{\int}_a^b f(x)dx = \overline{\int}_a^b f(x)dx.$$

It still remains to be shown that the definite integral defined above satisfies axioms (I0)-(I3). We turn to this task next.